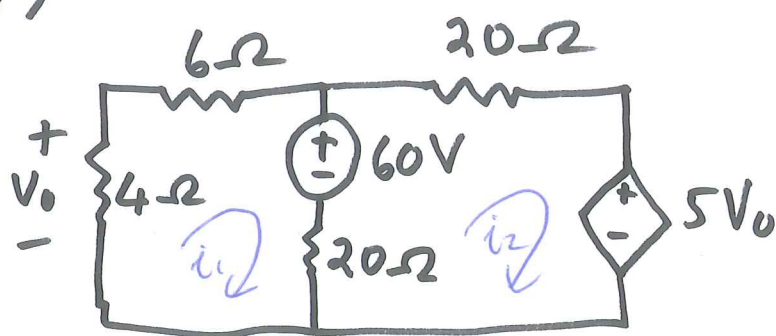


3.37)



Find V_o
Mesh

$$\text{M①: } -V_o + 6i_1 + 60 + 20(i_1 - i_2) = 0$$

$$30i_1 - 20i_2 = -60$$

$$15i_1 - 10i_2 = -30$$

$$\therefore i_2 = 1.5i_1 + 3 \dots \text{①}$$

$$\text{M② } 20(i_2 - i_1) - 60 + 20i_2 + 5V_o = 0$$

$$-40i_1 + 40i_2 = 60$$

$$-2i_1 + 2i_2 = 3 \dots \text{②}$$

sub ① in ②

$$-2i_1 + 2(1.5i_1 + 3) = 3$$

$$\therefore i_1 = -3 \text{ A} \dots \text{③}$$

sub ③ in ①

$$i_2 = 1.5(-3) + 3$$

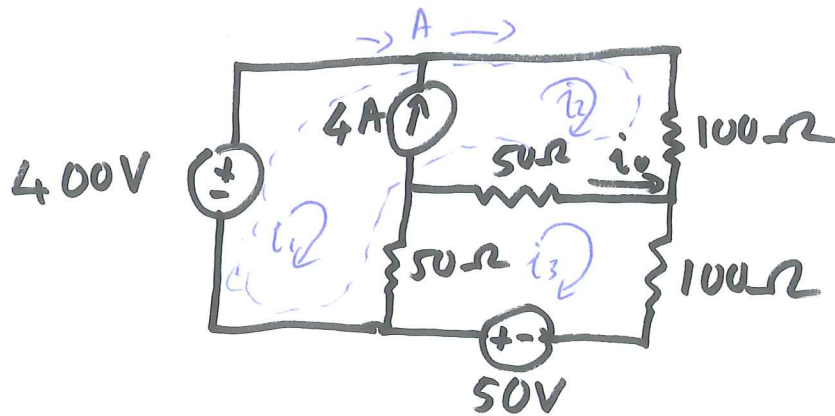
$$= -1.5 \text{ A}$$

$$V_o = -4(-3)$$

$$= 12 \text{ V}$$

→

3.44.



Find i_0

$$i_0 = i_3 - i_2$$

Super Mesh: $-400 + 100i_2 + 50i_2 - 50i_3 + 50i_1 - 50i_3 = 0$

$$50i_1 + 150i_2 - 100i_3 = 400$$

$$i_1 + 3i_2 - 2i_3 = 8 \quad \dots \textcircled{1}$$

Loop 3: $-50i_1 - 50i_2 + 200i_3 = 50$

$$-i_1 - i_2 + 4i_3 = 1 \quad \dots \textcircled{2}$$

Node A: $4 + i_1 - i_2 = 0$

$$i_1 = i_2 - 4 \quad \dots \textcircled{3}$$

Sub $\textcircled{3}$ in $\textcircled{1}$

$$i_2 - 4 + 3i_2 - 2i_3 = 8$$

$$4i_2 - 2i_3 = 12$$

$$2i_2 - i_3 = 6$$

$$i_3 = 2i_2 - 6 \quad \dots \textcircled{4}$$

Sub (3) in (2)

$$-i_2 + 4 - i_2 + 4i_3 = 1$$

$$-2i_2 + 4i_3 = -3 \dots (5)$$

Sub (4) in (5)

$$-2i_2 + 8i_2 - 24 = -3$$

$$6i_2 = 21$$

$$i_2 = 3,5 A \dots (6)$$

Sub (6) in (4)

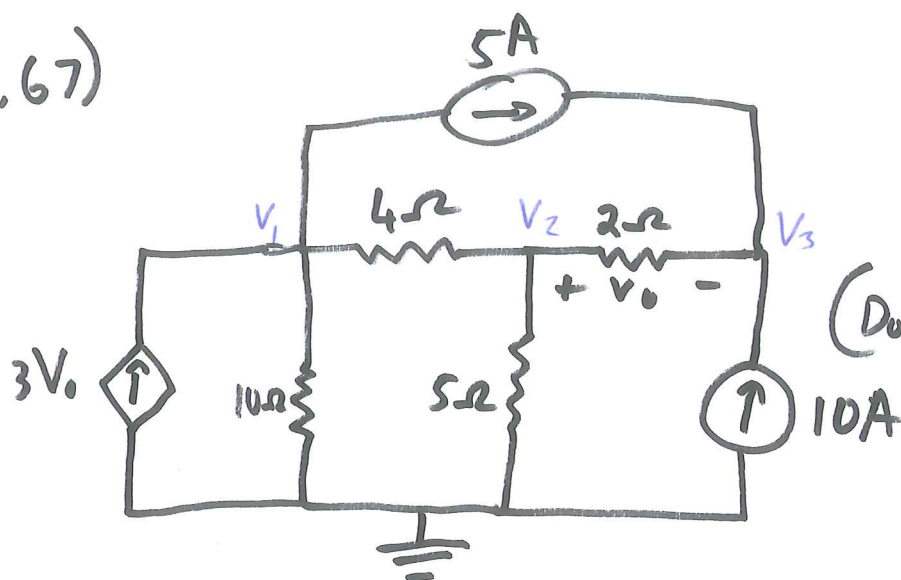
$$i_3 = 2(3,5) - 6 = 1 A$$

$$\therefore i_0 = i_3 - i_2$$

$$= 1 - 3,5$$

$$= -2,5 A$$

3.67)



Node Voltages eqn.
by Inspection

(Don't Find V_0)

Diagonal terms:

$$G_{11} = \frac{1}{10} + \frac{1}{4} = 0,35$$

$$G_{22} = \frac{1}{4} + \frac{1}{5} + \frac{1}{2} = 0,95$$

$$G_{33} = \frac{1}{2} = 0,5$$

Off diagonal

$$G_{12} = -\frac{1}{4} = -0,25$$

$$G_{13} = 0$$

$$G_{21} = -0,25$$

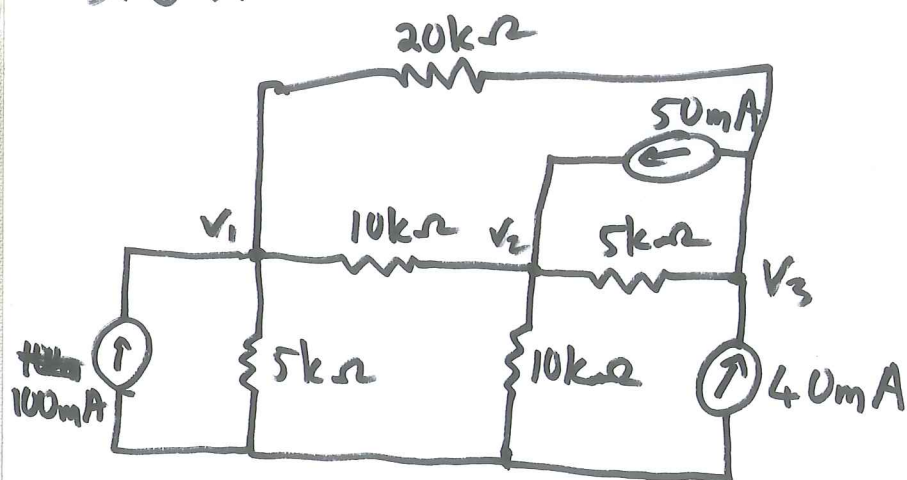
$$G_{23} = -0,5$$

$$G_{31} = 0$$

$$G_{32} = -0,5$$

$$\begin{bmatrix} 0,35 & -0,25 & 0 \\ -0,25 & 0,95 & -0,5 \\ 0 & -0,5 & 0,5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -5 + 3V_s \\ 0 \\ 15 \end{bmatrix}$$

3.69)



equ.
Node Voltages, by
inspection

Diagonal:

$$G_{11} = \left(\frac{1}{5} + \frac{1}{10} + \frac{1}{20} \right) = 0,35$$

$$G_{22} = \left(\frac{1}{10} + \frac{1}{10} + \frac{1}{5} \right) = 0,4$$

$$G_{33} = \left(\frac{1}{5} + \frac{1}{20} \right) = 0,25$$

off diagonal

$$G_{12} = -0,1 ; G_{13} = -0,05$$

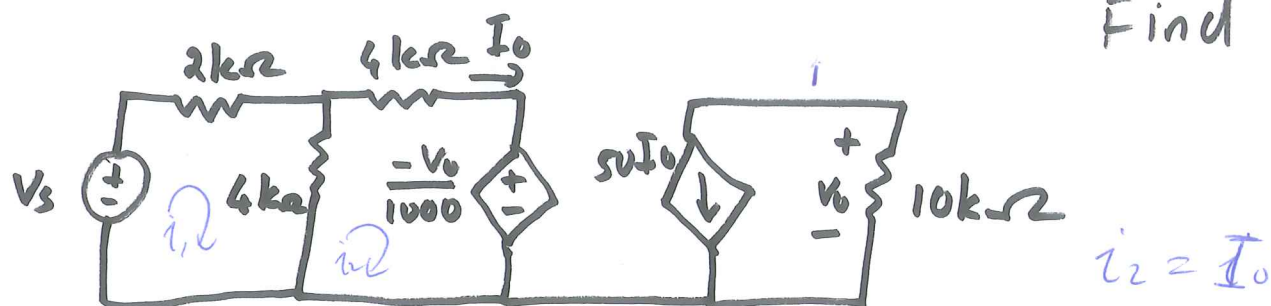
$$G_{21} = -0,1 ; G_{23} = -0,2$$

$$G_{31} = -0,05 ; G_{32} = -0,2$$

$$\begin{bmatrix} 0,35 & -0,1 & -0,05 \\ -0,1 & 0,4 & -0,2 \\ -0,05 & -0,2 & 0,25 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 100 \\ 50 \\ -10 \end{bmatrix}$$

3.88)

Find $\frac{V_o}{V_s}$



Relationship between V_o and I_o

$$\frac{0 - V_o}{10k} = 50 I_o$$

$$\therefore I_o = -\frac{V_o}{500k}$$

Loop 1: $6000 i_1 - 4000 i_2 = V_s \dots \textcircled{1}$

Loop 2: $-4000 i_1 + 8000 i_2 = \frac{V_o}{1000}$
 $-6000 i_1 + 12000 i_2 = \frac{3V_o}{2000} \dots \textcircled{2}$

$\textcircled{1} + \textcircled{2}$: $8000 i_2 = V_s + \frac{3V_o}{2000}$

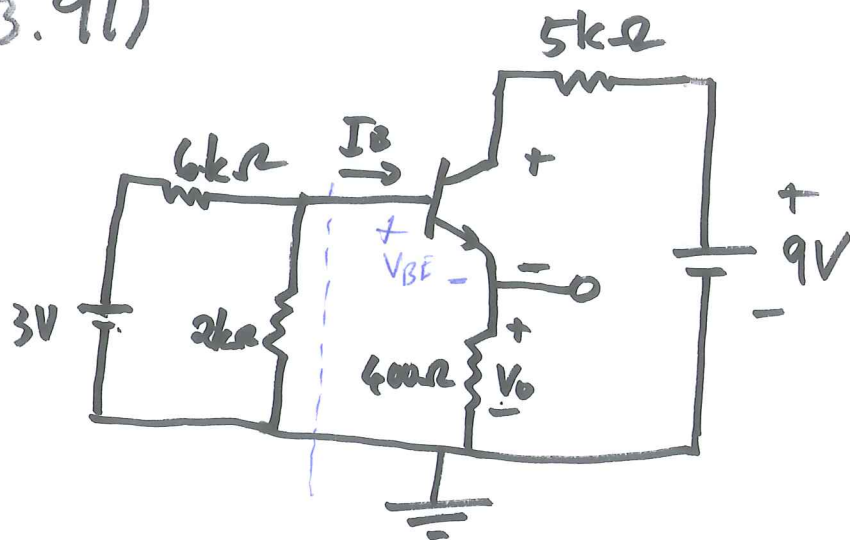
Sub i_o : $8000 i_o = V_s + \frac{3V_o}{2000}$

$$-\left(\frac{8000}{500000} + \frac{3}{2000}\right)V_o = V_s$$

$$\therefore \frac{V_o}{V_s} = -\frac{2000}{35}$$

$$= -57.14$$

3.91)



Find I_B , V_{CE} and V_o .

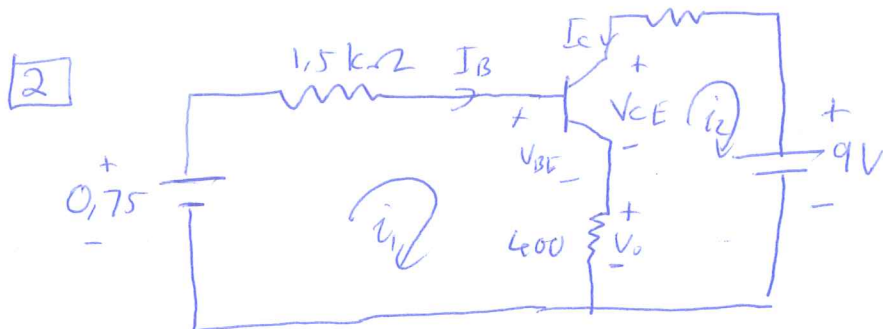
$\beta = 150$ and $V_{BE} = 0.7V$

~~Find~~

1) Determine the Thevenin equivalent circuit for the 3V source, $6k\Omega$ resistor and the $2k\Omega$ resistor

$$R_{th} = 6 \parallel 2 = 1.5k\Omega$$

$$V_{th} = V_{2k\Omega} = \left(\frac{2}{8}\right)(3) = 0.75V$$



3)

$$-0.75 + (1.5)(1000)I_B + 0.7 + 400I_E = 0$$

$$(-0.75 + 0.7) + 1500I_B + 400((1+\beta)I_B) = 0$$

$$I_B(1500 + 80400) = 0.05$$

$$I_B = 0.61\mu A$$

4 V_o :

$$R = \frac{V}{I}$$

$$G_{v0} = \frac{V_o}{I_E}$$

$$G_{v0} = \frac{V_o}{(1+\beta)I_B}$$

$$\therefore V_o = G_{v0} (201) (0,61 \times 10^{-6})$$

$$\underline{V_o = 49 \text{ mV}}$$

5 Mesh i_2 :

$$i_2 = -I_C = -(\beta)I_B$$

$$i_2 = -0,122 \text{ mA}$$

$$(5000)i_2 + 9 - V_o - V_{CE} = 0$$

$$(5000)(-0,122 \times 10^{-3}) + 9 - (0,049) - V_{CE} = 0$$

$$\therefore \underline{V_{CE} = 8,341 \text{ V}}$$

Answers

$$3.37) V_0 = 12V$$

$$3.41) i = 1,19A$$

$$3.44) i_0 = -2,5A$$

$$3.49) V_0 = 114V ; i_0 = 36A$$

$$3.67) \begin{bmatrix} 0,35 & -0,25 & 0 \\ -0,25 & 0,45 & -0,5 \\ 0 & -0,5 & 0,5 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -5 + 3V_0 \\ 0 \\ 15 \end{bmatrix}$$

$$3.69) \begin{bmatrix} 0,35 & -0,1 & -0,05 \\ -0,1 & 0,4 & -0,2 \\ -0,05 & -0,2 & 0,25 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 100 \\ 50 \\ -10 \end{bmatrix}$$

$$3.70) V_1 = -4V ; V_2 = 5V$$

$$3.72) \begin{bmatrix} 30 & 0 & -10 & 0 \\ 0 & 50 & 0 & -30 \\ -10 & 0 & 30 & -20 \\ 0 & -30 & -20 & 50 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 40 \\ -15 \\ 15 \\ -20 \end{bmatrix}$$

$$3.84) V_0 = 2,5V ; I_0 = 2,5\mu A$$

$$3.86) V_o = -117,06 \text{ mV}$$

$$3.88) \frac{V_o}{V_s} = -57,14$$

$$3.91) I_B = 0,61 \mu\text{A}$$

$$V_{CE} = 8,341 \text{ V}$$

$$V_o = 49 \text{ mV}$$